

# KATRA TEJASRI

## Design Verification Engineer

SystemVerilog ■ UVM ■ AXI4 ■ Ethernet MAC ■ MIPI I3C

katra.tejasri31@gmail.com | +91 90636847112 | Hyderabad, India

linkedin.com/in/katra-tejasri-412891141

## PROFESSIONAL SUMMARY

Design Verification Engineer with 2.6 years of experience in IP-level functional verification using SystemVerilog and UVM methodology. Experienced in verifying high-speed protocols including Ethernet MAC, AXI4, MIPI I3C, APB, SPI, and I2C. Skilled in developing reusable UVM environments, assertions, scoreboards, constrained-random testbenches, and functional coverage models. Strong debugging expertise using Synopsys VCS and Verdi.

## TECHNICAL SKILLS

|                            |                                                                                       |
|----------------------------|---------------------------------------------------------------------------------------|
| <b>Languages</b>           | SystemVerilog, Verilog, Python                                                        |
| <b>Methodologies</b>       | UVM, Constrained Random Verification, Functional Coverage, SVA                        |
| <b>Protocols</b>           | Ethernet MAC, AXI4, MIPI I3C                                                          |
| <b>Verification Skills</b> | UVM TB Architecture, Assertions, Scoreboards, Coverage Closure, Debugging, Regression |
| <b>EDA Tools</b>           | Synopsys VCS, Verdi, QuestaSim                                                        |

## PROFESSIONAL EXPERIENCE

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>Design Verification Engineer</b><br>Miliueidigital Technologies Pvt Ltd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | June 2024 – Present |
| <b>Ethernet MAC Subsystem Verification</b><br><b>Client:</b> SI-LABS<br><b>Environment:</b> SystemVerilog, UVM, Synopsys VCS, Verdi<br><b>Protocols:</b> Ethernet (CDMII), AXI4-Stream, APB <ul style="list-style-type: none"><li>Developed reusable UVM agents for CDMII, AXI4-Stream, and APB interfaces.</li><li>Implemented Ethernet frame generation including preamble, SFD, VLAN, payload, and CRC32.</li><li>Designed scoreboard and analysis FIFOs for end-to-end frame checking.</li><li>Developed AXI and APB master verification components for MAC configuration and data movement.</li><li>Executed constrained-random verification for MAC datapath validation.</li><li>Debugged protocol and data integrity issues using Verdi waveform analysis.</li></ul> | Sep 2025 – Present  |
| <b>I3C Protocol IP Verification</b><br><b>Client:</b> Hexasemi Solutions<br><b>Environment:</b> SystemVerilog, UVM, Synopsys VCS, Verdi<br><b>Protocol:</b> MIPI I3C <ul style="list-style-type: none"><li>Developed UVM-based verification environment for MIPI I3C protocol IP.</li><li>Implemented sequences and constraints for SDR transfers, CCC commands, and dynamic address assignment.</li><li>Verified directed/broadcast CCC operations and I2C backward compatibility.</li><li>Developed functional coverage for protocol feature validation.</li><li>Debugged RTL and testbench issues using simulation logs and waveform analysis.</li></ul>                                                                                                                 | Dec 2024 – Aug 2025 |
| <b>AXI4 IP Verification</b><br><b>Client:</b> Agastya Labs<br><b>Environment:</b> SystemVerilog, UVM, Synopsys VCS, Verdi<br><b>Protocol:</b> AMBA AXI4 <ul style="list-style-type: none"><li>Developed UVM verification components for AXI4 protocol validation.</li><li>Verified burst transfers, handshake behavior, out-of-order, and interleaved transactions.</li><li>Implemented SVA assertions for protocol rule checking.</li><li>Performed coverage analysis and regression debugging for verification closure.</li><li>Developed transaction-level scoreboards and protocol monitors.</li></ul>                                                                                                                                                                  | Jun 2024 – Nov 2024 |

## INTERNSHIP

|                                                                                                                                                                                                                                                         |                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>Future Wireless Communication (FWC) Intern</b><br>Indian Institute of Technology Hyderabad                                                                                                                                                           | Oct 2023 – Mar 2024 |
| <ul style="list-style-type: none"><li>Worked on protocol-level IP verification and simulation activities.</li><li>Assisted in verification environment setup and debugging.</li><li>Participated in waveform analysis and testcase execution.</li></ul> |                     |

## EDUCATION

| Degree        | Institution                             | Score     | Year |
|---------------|-----------------------------------------|-----------|------|
| M.Tech – VLSI | JBLET (JNTUH), Hyderabad                | CGPA: 9.2 | 2022 |
| B.Tech – EEE  | Anubose Institute of Technology (JNTUH) | 70%       | 2017 |

## KEY HIGHLIGHTS

- 2.6 years of hands-on ASIC/IP verification experience.
- Strong expertise in SystemVerilog and UVM-based verification methodologies.
- Experience with Ethernet MAC, AXI4, and MIPI I3C protocols.
- Skilled in assertions, scoreboards, functional coverage, and debug analysis.
- Available for Remote and Hybrid opportunities.

## CERTIFICATIONS

- Future Wireless Communication (FWC) Internship Certificate – Indian Institute of Technology Hyderabad (IITH)